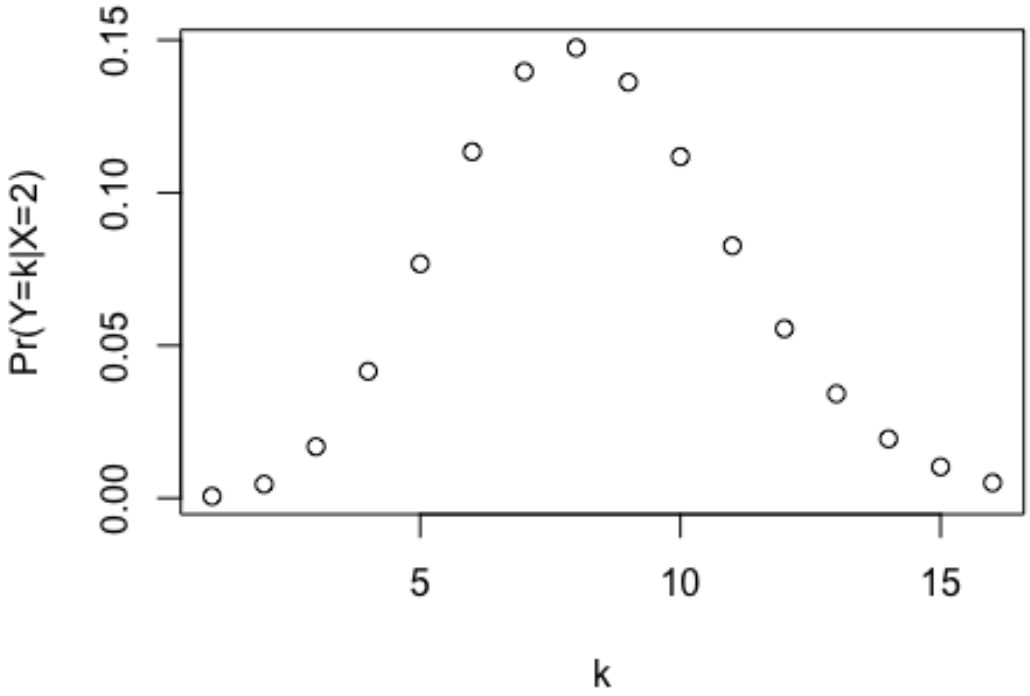
Problem Set 5 by Jimmy Zheng

2023-02-08

Question 1

(a) Plot Pr(Y=k|X=2) for k=0 ,1,...,15.

```
lambda = exp(1 + 0.5*2)
plot(dpois(0:15, lambda = lambda), ylab = "Pr(Y=k|X=2)", xlab = "k")
```



(b) What is the most plausible value for Y when X=2 ?

```
which.max(dpois(0:15, lambda = lambda))
```

```
## [1] 8
```

(c) Construct a 95% prediction interval for Y.

```
qpois(c(0.025, 0.975), lambda = lambda)
```

```
## [1]  3 13
```

Question 2: ISL Chapter 4: Exercise (applied) 13(e)-(i).

```
# Import the data
library(ISLR2)

# Split the Dataset
train = (Weekly$Year < 2009)
Weekly_09_to_10 = Weekly[!train, ]

# The Logistic Regression Model
mod_log = glm(Direction ~ Lag2, data = Weekly, family = binomial, subset =
train)

# Confusion Matrix and Overall Error Rate
predicted_prob_log = predict(mod_log, Weekly_09_to_10, type = "response")
predicted_direction_log = rep("Down", length(predicted_prob_log))
predicted_direction_log[predicted_prob_log > 0.5] = "Up"
Direction_09_to_10 = Weekly$Direction[!train]

table(predicted_direction_log, Direction_09_to_10)
```

```
##                         Direction_09_to_10
## predicted_direction_log Down Up
##                    Down    9  5
##                    Up     34 56
```

```
mean(predicted_direction_log==Direction_09_to_10)
```

```
## [1] 0.625
```

(e) Repeat (d) using LDA.

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:ISLR2':
##
##     Boston
```

```
mod_lda = lda(Direction ~ Lag2, data = Weekly, subset = train)

# Confusion Matrix and Overall Correct Prediction Rate
lda.pred = predict(mod_lda, Weekly_09_to_10)
lda.class = lda.pred$class
table(lda.class, Direction_09_to_10)
```

```
##          Direction_09_to_10
## lda.class Down Up
##      Down    9  5
##      Up     34 56
```

```
mean(lda.class==Direction_09_to_10)
```

```
## [1] 0.625
```

(f) Repeat (d) using QDA.

```
mod_qda = qda(Direction ~ Lag2, data = Weekly, subset = train)

# Confusion Matrix and Overall Correct Prediction Rate
qda.class = predict(mod_qda, Weekly_09_to_10)$class
table(qda.class, Direction_09_to_10)
```

```
##          Direction_09_to_10
## qda.class Down Up
##      Down    0  0
##      Up     43 61
```

```
mean(qda.class==Direction_09_to_10)
```

```
## [1] 0.5865385
```

(g) Repeat (d) using KNN with K = 1.

```
library(class)
set.seed(1)
train.X = Weekly[train, ]$Lag2
test.X = Weekly[!train, ]$Lag2
train.Direction = Weekly[train, ]$Direction

# make sure train.X and test.X has the same dimension
dim(train.X) = c(length(train.X), 1)
dim(test.X) = c(length(test.X), 1)

knn.pred = knn(train.X, test.X, train.Direction, k=1)

# Confusion Matrix and Overall Correct Prediction Rate
table(knn.pred, Direction_09_to_10)
```

```
##         Direction_09_to_10
## knn.pred Down Up
##     Down   21 30
##     Up     22 31
```

```
mean(knn.pred==Direction_09_to_10)
```

```
## [1] 0.5
```

(h) Repeat (d) using naive Bayes.

```
library(e1071)
mod_nb = naiveBayes(Direction ~ Lag2, data = Weekly, subset = train)

# Confusion Matrix and Overall Correct Prediction Rate
```

```
nb.class = predict(mod_nb, Weekly_09_to_10)
table(nb.class, Direction_09_to_10)

##         Direction_09_to_10
## nb.class Down Up
##     Down    0  0
##     Up     43 61

mean(nb.class==Direction_09_to_10)

## [1] 0.5865385
```

(i) Which of these methods appears to provide the best results on this data?
The logistic regression model and LDA provide the best results since they have the highest correct prediction rate of 0.625.

Question 3: ISL Chapter 5: Exercise 1.

1. Using basic statistical properties of the variance, as well as single variable calculus, derive (5.6). In other words, prove that α given by (5.6) does indeed minimize Var(aX + (1 − a)Y).

Denote $\sigma_X^2 = Var(X)$, $\sigma_Y^2 = Var(Y)$, $\sigma_{XY} = Cov(X,Y)$

$$Var(\alpha X + (1-\alpha)Y) = \alpha^2 Var(X) + (1-\alpha)^2 Var(Y) + 2\alpha(1-\alpha)Cov(X,Y)$$

$$= \alpha^2\sigma_X^2 + (1-2\alpha+\alpha^2)\sigma_Y^2 + (2\alpha - 2\alpha^2)\sigma_{XY}$$

Denote $f(\alpha) = \alpha^2\sigma_X^2 + (1-2\alpha+\alpha^2)\sigma_Y^2 + (2\alpha-2\alpha^2)\sigma_{XY}$

$$\frac{\partial f(\alpha)}{\partial \alpha} = 2\alpha\sigma_X^2 - 2\sigma_Y^2 + 2\alpha\sigma_Y^2 + 2\sigma_{XY} - 4\alpha\sigma_{XY}$$

Set $\frac{\partial f(\alpha)}{\partial \alpha} = 0$:

$$2\alpha\sigma_X^2 - 2\sigma_Y^2 + 2\alpha\sigma_Y^2 + 2\sigma_{XY} - 4\alpha\sigma_{XY} = 0$$

$$\alpha\sigma_X^2 + \alpha\sigma_Y^2 - \alpha\sigma_{XY} - \sigma_Y^2 + \sigma_{XY} = 0$$

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - \sigma_{XY}}$$

//